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1. Introduction

This report summarizes the work completed during the internship, which focused on studying dimensionality reduction methods such as Principal Component Analysis (PCA) and their nonlinear extensions through the application of tools from topological data analysis (TDA). The main goal of the practice was to develop skills in data analysis, Python programming, and experimental research with synthetic datasets.

A special emphasis was placed on the use of topological methods for analyzing simple geometric objects. Unlike classical approaches, which rely mainly on geometric distances or statistical characteristics, TDA makes it possible to capture global properties of data, such as connected components, cycles, and higher-dimensional holes. This allows one to reveal the hidden structure and shape of a point cloud, even in the presence of noise.

To validate the chosen methodology, synthetic point samples on a circle and a torus were studied, as these shapes have well-known topological features that help verify the correctness of algorithms. The step-by-step construction of simplicial complexes and the calculation of categorical characteristics made it possible to trace structural changes under different parameter settings and to demonstrate the robustness and effectiveness of the method in noisy data conditions.

2. Object of the Internship

The main objective of the internship was to study Principal Component Analysis (PCA) as a classical linear dimensionality reduction method, identify its limitations on nonlinear datasets, and explore topological approaches for developing nonlinear analogs of PCA. To achieve this, synthetic datasets based on classical geometric figures — a circle and a torus — were used. These shapes were chosen because of their clarity and well-defined topological properties: the circle has a single cycle, while the torus contains two independent cycles, making them especially suitable for testing the correctness of topological analysis algorithms.

The datasets were generated parametrically: for the circle, points were placed in the plane and then perturbed with random noise; for the torus, two angular parameters were used to define point positions in three-dimensional space, followed by the addition of noise. This modeling approach not only allowed testing under controlled conditions but also brought the data closer to real-world scenarios, where measurements are rarely perfectly accurate. Such settings made it possible to observe how the algorithms recover key topological characteristics under perturbations and distortions, thereby assessing the robustness of the topological methods compared to classical PCA.

3. Purpose of the Internship

The assignment focused on the implementation and verification of an algorithm for estimating the LS-category on sequences of Vietoris–Rips complexes constructed from noisy point clouds. The work involved generating synthetic datasets on circles and a torus with varying levels of noise, followed by the step-by-step construction of simplicial complexes for different values of the parameter ε and the computation of categorical characteristics. A Σ-matrix reflecting relationships between complexes was then formed, and spectral analysis of similarity matrices was applied to select the critical ε.

Special attention was devoted to noise handling: isolated points that distort the topological structure were identified and removed, after which the corrected results were compared to the original ones. This made it possible to evaluate the influence of local complexity bounds on the final estimation. Additionally, the robustness of the algorithm was tested against noise intensity, sample size, and parameter discretization, while visualization of results allowed tracing structural changes. Altogether, these steps demonstrated the method's potential for extracting stable topological features and analyzing data behavior under different experimental conditions.

4. Description of Explored Medium

As a test environment for the implementation, synthetic point clouds were generated that imitate classical geometric objects with added noise. Two main shapes were considered: a circle in a two-dimensional space and a torus in a three-dimensional space. The use of artificial data made it possible to control the number of points and the level of noise, which in turn provided clarity in analyzing the effect of parameters on the behavior of complexes **(Figure 4.1)**.

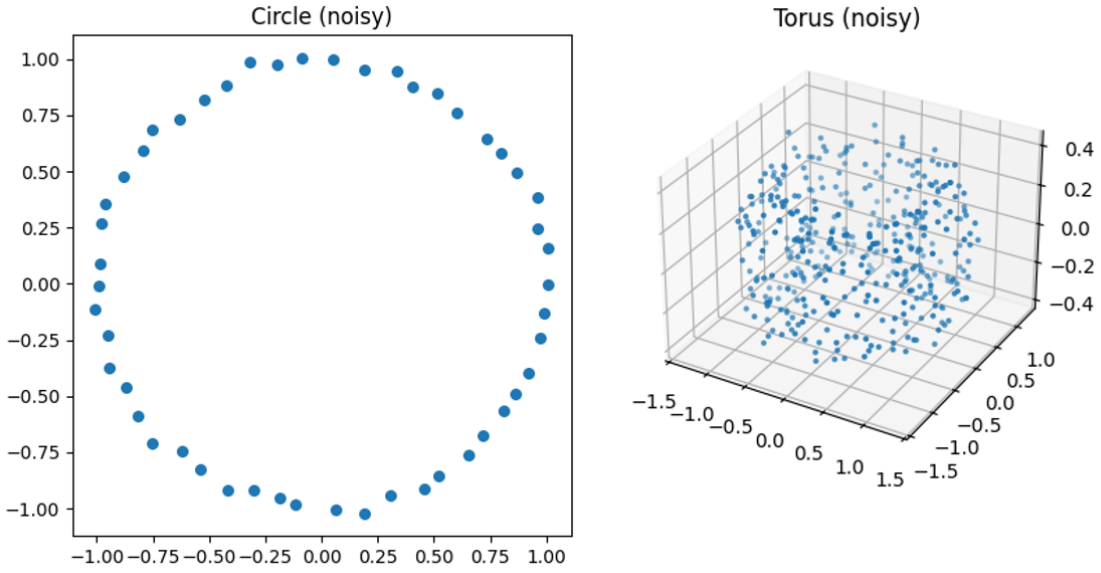


Figure 4.1. Examples of generated data: circle and torus with noise

For the circle, 50 points were distributed along the unit circumference with a small Gaussian noise added to their coordinates. This allowed the dataset to retain a recognizable circular structure while still containing random deviations that complicate topological analysis. For the torus, 400 points were generated on the surface using a parametric representation with radii 1.0 and 0.3. Random noise was also introduced into this construction, which created a realistic but less regular point distribution in three dimensions. These datasets were chosen as they represent two types of topological objects: a one-dimensional loop (circle) and a surface with non-trivial higher-order homology (torus). This combination made it possible to check how persistent homology methods and LS-category behave both on simple and more complex structures.

During the work, Vietoris–Rips complexes were built on these point clouds at different scales ε. This allowed tracking how the connectivity of the structures changes with increasing ε: from isolated vertices to a fully connected graph and higher-dimensional simplices. Such an environment turned out to be convenient for demonstrating the transition from purely combinatorial characteristics to topological invariants.

The work was carried out in the Google Colab cloud environment, which provided access to computational resources and simplified the process of testing the code. Using Colab allowed interactive work: running code cells, checking intermediate results, adjusting parameters, and instantly visualizing changes. Python was chosen as the main programming language, since it provides a wide range of libraries covering data generation, numerical computations, topological analysis, and visualization. For data generation and numerical processing, the NumPy library was used, which allows efficient work with multidimensional arrays and vectorized operations. Additionally, the SciPy library was applied, particularly its module for computing distance matrices, which was essential for building Vietoris–Rips complexes. Data visualization was performed using Matplotlib, which enabled the creation of two- and three-dimensional plots of point clouds, LS-category curves, and persistence diagrams. For linear algebra tasks and eigenvalue analysis, functions from SciPy.linalg were applied, allowing calculation of the Σ-matrix spectrum and identification of the principal eigenvector. The key tool was the GUDHI library, which provides ready-to-use implementations for topological data analysis. With its help, Vietoris–Rips complexes were built, simplex trees were formed, and persistence diagrams for different dimensions were computed. Another important library was scikit-learn, from which PCA was used to reduce dimensionality and visually compare original and cleaned datasets. Altogether, these tools formed a working environment where the entire algorithm could be implemented: from point generation and noise addition to the construction of topological invariants and their visualization.

Thus, this toolset made it possible to carry out the full workflow, including data preparation, algorithm implementation, calculation of characteristics, and their interpretation.

5. Theoretical part

a) For the analysis of a set of points X in a metric space, we define the Vietoris–Rips complex $VR_\varepsilon(x)$ for $a \le b, M = \frac{diam(X)}{2}$

When ε increases, the inclusion $VR_a(X) \subseteq VR_b(X), a \le b$ holds. That is, for the small ε the complex consists of separate vertices, for large – it tends to a complete simplex. We consider intermediate values, where the topological shape (circle, torus) is revealed.

b) The Lusternik–Schnirelmann category (LS-category) measures the "topological complexity" of an object.

$cat(K)$ - the Lusternik–Schnirelmann category: the minimal number of open contractible sets covering K.

For two complexes K_1, K_2, their union is denoted as $K_1 \oplus K_2 = \{s \mid s \in K_1 \, ors \in K_2\}$. Then we define the "scalar product" $< K_1, K_2 >= -cat(K_1 \oplus K_2) + cat(K_1) + cat(K_2)$.

Then the matrix $\Sigma_N = \left(< VR_{\varepsilon_i}(X), VR_{\varepsilon_j}(X) > \right)_{i,j=1}^{N}$ is constructed. The matrix Σ is symmetric and positive. By the Frobenius–Perron theorem it has the largest eigenvalue λ_N^* with the eigenvector v_N^*, all its coordinates are positive.

c) Since direct computation of the LS-category is laborious, the point complexity of a vertex x is introduced.

$pc(x) = d - max\{dim(\sigma) \mid x \in \sigma\}$. Then, $cat(VR_\varepsilon(X)) \le \sum_{x \in X} pc(x)$. This estimate is computed quickly and allows comparing different scales of ε.

Using spectral analysis of the matrix Σ we find the principal eigenvector v^*. The critical value ε is chosen by the coordinate with the largest weight: $\varepsilon_{crit} = \varepsilon_i$, $i = arg\ max\{v_j(k)\}$. ε_{crit} corresponds to the "optimal scale" where the topological structure of the data is revealed most clearly.

e) For $\varepsilon = \varepsilon_{crit}$ isolated vertices (those that are not connected with others in the 1-skeleton) are removed from the complex. $X' = X \setminus \{x \in X \mid x \; is \; isolated \; in \; VR_{\varepsilon_{crit}}(X)\}$. Accordingly, isolated vertices are most often noise and distort the topological picture.

6. Description of Achieved Results

We investigate two types of nonlinear objects:

Circle S^1 and torus T^2

$$S^1 = \{(cos\ t, sin\ t)\},$$

$$T^2 = \{((R + r\ cos\ \theta)cos\ \phi, (R + r\ cos\ \theta)sin\ \phi, r\ sin\ \theta)\}$$

Adding noise.

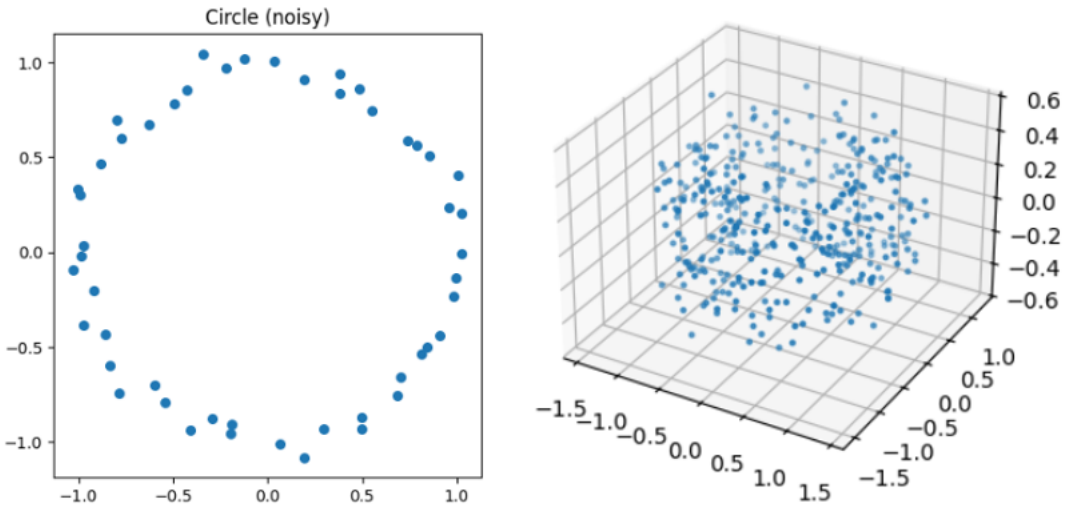


Figure 6.1. Examples of noisy data (circle and torus)

Considering factors:

1. **Noise level**: small noise ($\sigma = 0.02\ /\ 0.05$), medium noise ($\sigma = 0.05\ /\ 0.1$), strong noise ($\sigma = 0.1\ /\ 0.15$)
2. **Sample size**
3. **Allowed grids over** ε

Depending on the variability of the noise, in the circle with small and medium noise there are no isolated points. With strong noise 1 isolated point appeared (2%). Consequently, the algorithm is resistant to noise on a small sample until the noise becomes very large.

Noise level

Torus: with small noise – 8 isolated points (2%), with medium noise – 29 isolated points (~7%), with strong noise – 14 isolated points (~3.5%). In this case the number of isolated points does not grow monotonically, but depends on the "geometry of the noise" and the chosen ε_{crit}. But there is a trend: as the noise increases the probability of isolated points also increases.

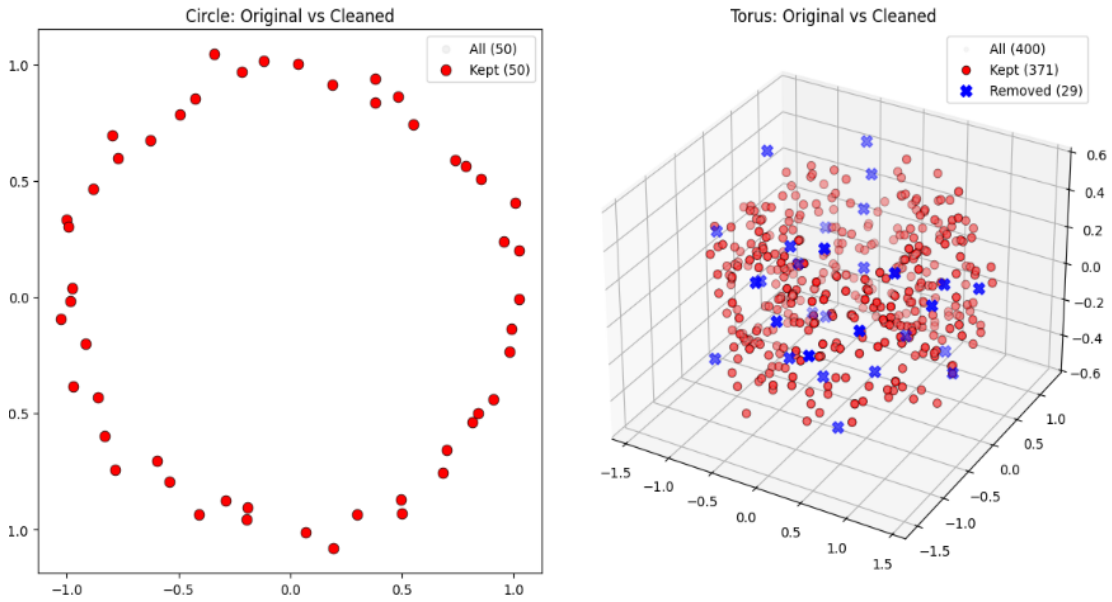


Figure 6.2. Comparison of original and cleaned samples

Influence of the sample size on the number of isolated points and on ε_{crit}

With a small sample **(n=50)** in the topological structure fluctuates, artifacts are possible. With a large sample, for example **n=200**, the structure becomes more stable: more edges and simplices, the LS-category smooths out, the critical ε stabilizes. This is an analogue of the "convergence" of the method with the growth of data

Influence of the ε. (num_epsilons = 15/30/60)

With a small grid **(15 steps)** the search for the critical ε is rough. The principal vector v* "jumps", the index of the chosen ε can depend on the discretization. With a medium grid **(30 steps)** the graphs are stable, the computations take reasonable time, while with a very detailed grid **(60 steps)** the result is smoother, but the computation time grows.

For each complex, the LS-category was computed, taking into account the contribution of all vertices through their point complexity. This allowed plotting the curves cat(VR_ε) and showing how the category estimate changes with increasing ε **(Figure 6.3)**.

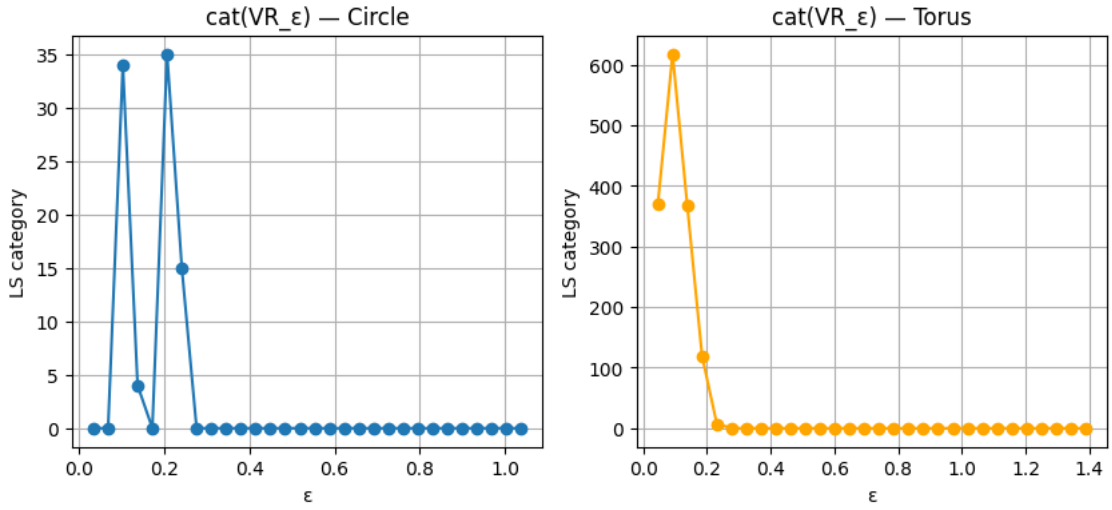


Figure 6.3. Dynamics of the LS-category for the circle and the torus as the parameter ε varies

Additionally, Σ-matrices were obtained, from which eigenvalues and eigenvectors were calculated. Determining the principal eigenvector made it possible to identify a critical ε at which the structure of the complex reflects the key topological features of the original data **(Figure 6.4)**. At this stage, isolated vertices were identified, and their removal showed that cleaned samples better preserved global topological features.

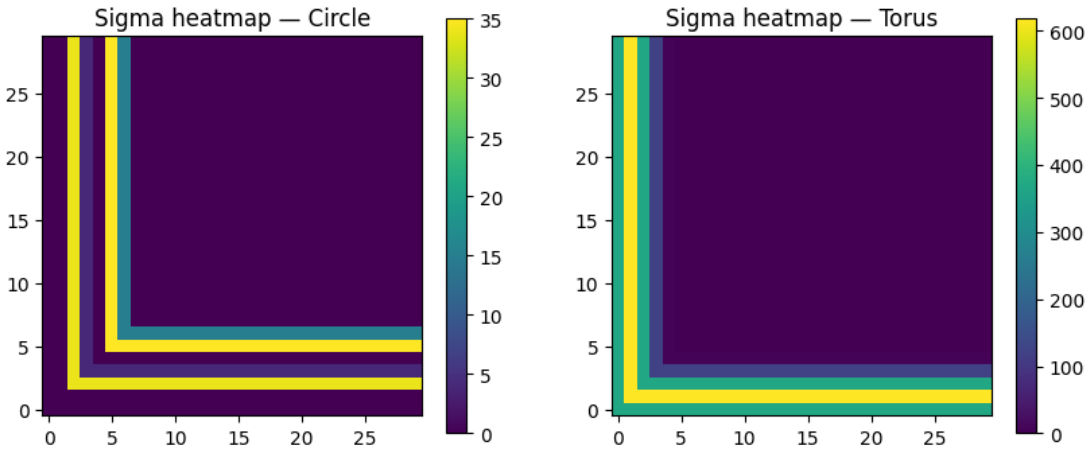


Figure 6.4. Heatmaps of the Σ-matrix for the circle and the torus

Special attention was paid to comparing original and corrected LS-category values. For this, local complexity bounds were introduced, which adjusted point complexity if a vertex had sufficient coverage by edges and triangles. As a result, new category curves were obtained, more robust to noise. Experiments with different levels of σ demonstrated that corrected estimates better reflected the real structure. Persistence diagrams for H_1 and H_2 were also built, clearly showing cycles and two-dimensional holes before and after cleaning **(Figure 6.5)**. Finally, analysis of the Σ spectrum and Ritz curves revealed differences between original and corrected matrices, confirming the effect of local complexity bounds on the final results. Consequently, a comprehensive picture of LS-category behavior and its relation to data topology was obtained.

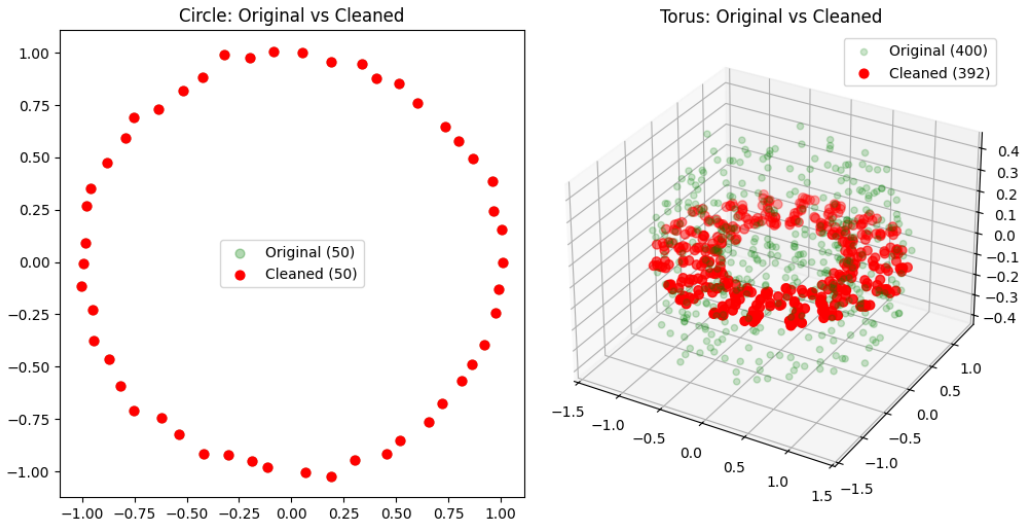


Figure 6.5. Comparison of original and cleaned data

7. Conclusion

During the industrial practice, an algorithm was implemented to study the connection between the behavior of Vietoris–Rips complexes and categorical characteristics of data. Using synthetic datasets on a circle and a torus with different noise levels, sequences of complexes were constructed for varying ε values, demonstrating how edges, triangles, and higher-dimensional structures emerge. For each complex, the LS-category was calculated, which made it possible to trace characteristic changes in the cat(VR_ε) function. Based on these results, Σ-matrices were formed, and their spectral analysis was used to determine a critical ε that reflects the key topological features of the dataset.

Special attention was given to noise handling: the removal of isolated points preserved the main topological structure and led to more stable results. Corrections introduced through local complexity bounds reduced sensitivity to noise, while comparison of original and corrected LS-category curves showed that both soft and strict corrections smooth out abrupt jumps and better capture the actual shape of objects. Persistence diagrams for H_1 and H_2 confirmed the presence of cycles and two-dimensional holes and illustrated how these features change after filtering. Analysis of Σ-matrix spectra and the evolution of the principal eigenvector further demonstrated the robustness of topological characteristics under parameter variation.

Overall, the study showed how classical PCA can be extended to nonlinear cases using topological methods. The combined application of LS-category, local complexity bounds, and persistent homology enabled the extraction of stable topological features even under significant distortions, confirming the potential of topology-based methods as a powerful alternative to PCA when analyzing complex nonlinear datasets.

The code of the work can be found by link: https://github.com/ek-lazareva/Internship-2025/blob/main/Interinship.ipynb

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